

Behavior of Engineered Cementitious Composites under Sulphate and Chloride Exposure

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ABSTRACT

A new generation of high-performance Fiber-reinforced cementitious composites with moderate Fiber content and good ductility is known as Engineered Cementitious Composites (ECC). The durability of these materials is crucial for ensuring the sustainability of concrete infrastructure. A durability performance of ECC with Fly Ash and Ground Granulated Blast Furnace Slag (GGBS) as supplementary cementitious materials was assessed. Experimental Program investigated the compressive and flexural strength behaviour of composites under continued exposure to sulphate and chloride environments for about 400 days. Microstructure of ECC subjected to those environmental conditions were examined under Scanning Electron Microscope (SEM). The results demonstrated that ECC exhibited better resistance to sulphate and chloride attack by also maintaining its structural integrity and mechanical performance even after prolonged exposure. Equations generated through curve fitting method to verify the experimental findings and R^2 value close to one represents a strong concordance between predicted and experimental values. Further, microstructural investigation revealed that hydration products, matrix toughness, and matrix-fiber bonding before and after susceptibility assessment in the sulphate and chloride dominated environments.

Keywords: ECC; GGBS; Durability; Sulphate; Chloride; Curve Fitting; SEM.

1. Introduction

Concrete, the major structural material in construction, poses substantial societal, economic, and environmental outcomes. Cement manufacturing causes approximately 5% of total global greenhouse gas emissions, while emitting harmful pollutants, including NO_x and particulate matter, into the air. The rising demand for sustainable construction practices and green building technologies over the past decade has stimulated a critical reassessment of construction methodologies and material selection [1]. Sustainability initiatives around the world have established goals for developing eco-friendly materials, energy-efficient design and construction methods that minimizes the environmental footprint and waste generation. ECC represents a

paradigm-changing advancement in the field of material engineering as a breakthrough technology [2-4].

ECC, also known as Strain Hardening Cement-Based Composite or Blended Concrete [5], is a unique mortar-based composite that incorporates specially selected short random polymer fibers as reinforcement, distinguishing it from regular concrete. This composite serves to develop a durable and flexible material suitable for resolving difficulties present in the traditional Fiber-Reinforced Concrete. Increase in the compressive strength tend to result in brittle behavior of materials thus restricting high-strength concrete applications for particular structures. This limitation is surpassed in ECC as it provides high strength and also enhanced tensile deformation (strain) capabilities than conventional and other fiber-reinforced concrete. The strain capacity of ECC ranging from 3% to 7%, whereas Ordinary Portland Cement (OPC) displays only 0.01% [3,5].

The capability of cement concrete to withstand weathering, chemical attacks, abrasion, and other deterioration processes is referred to as durability. A durable concrete structure maintains its quality, and functionality even when exposed to various environmental conditions and external factors. It is a critical characteristic that ensures the long-term performance and serviceability of concrete in its intended application [6]. Sulphate attack is one such significant durability concerns, particularly affecting hydraulic structures. When hydrated cement products reacts with sulphate ions forms an expansive compound, which causes cracking in the concrete. As these cracks allows sulphate ions to penetrate into concrete which intensify the deterioration process. Further the hydraulic structures in sulphate rich environment are at high probability of failure, leading to economic losses and societal impact [6]. Conversely, the entry of chloride ions in to concrete can cause corrosion, so controlling the cracks is so essential to minimize the chloride transport rate in composites [7].

The work mainly aims to develop ECC using industrial byproducts which significantly contributes to the ecofriendly construction. Using an industrial waste as alternative for cement resolves many environmental issues associated with the construction industry combined with resource efficient construction. Further the study examines the long-term exposure of composites under sulphate and chloride condition which gives an insight about the performance of composite in chemical degrading environment. Finally, scientific findings of this study on performance of ECC in these conditions offering a practical solution for construction

professionals and engineers who are aiming to enhancing the structural durability of buildings by also keeping sustainability practices in construction.

2. Experimental Procedure

2.1 Materials and Methods

The ECC mixture comprises Ordinary Portland Cement [8], Class F Fly Ash [9], Ground Granulated Blast Furnace Slag (GGBS) [10], Crushed Sand (S) [11], Polypropylene Fibers (PPF), water, and BSF 8154 High-Range Water-Reducing Super plasticizers (HRWRSP). Chemical composition of Cement and cementitious materials were given in Table 1 and Table 2 shows the properties of PPF. The mix design adopted in this study is based on established literature [12,13], with the final composition achieved after successive laboratory trials. ECC mix ratios consisting of cementitious material to cement ratio (CM/C) of 1.2. The water to cementitious material ratio (W/CM) and crushed sand to cementitious material ratio (S/CM) are fixed at 0.28 and 0.36. Consistent with literature recommendations [12,13], the ECC mix design parameters are listed in Table 3.

Table 1. Chemical Composition of Cement, Fly Ash and GGBS.

Constituents	Chemical composition (% by mass)							
	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	SO ₃	CaO	MgO	Na ₂ O	LOI
Cement	20.98	5.42	3.92	2.36	62.85	1.76	0.28	1.90
FA	55.30	25.80	5.50	0.30	2.90	0.80	-	3.2
GGBS	34.06	20	0.80	0.90	32.60	7.89	-	3.72

Table 2. Characteristics of Polypropylene Fibers (PPF) used.

Length (mm)	Diameter (mm)	Density (g/cm ³)	Tensile strength (MPa)	Modulus of Elasticity (GPa)
12	0.05	0.91/cm ³	≥ 300	3.5

Table 3. Mix Proportions of ECC

Materials	Proportions
FA/C	0.8
GGBS/C	0.4

W/CM	0.28
S/CM	0.36
Water, kg/m ³	351
Cement (C), kg/m ³	570
Fly Ash (FA), kg/m ³	456
GGBS, kg/m ³	228
C-Sand (S), kg/m ³	452
PPF, kg/m ³	20
HRWRSP (%)	1.4

2.2 Mini-Slump Cone Test

Mini slump cone test was performed to evaluate the consistency of ECC mix. The bottom and top diameter of the cone are 80 mm and 70 mm respectively and height of cone is 40 mm [14]. The cone was placed on the square glass plate at its centre, then it is filled with ECC mix and lift the mould, and final spread diameter were measured. The average spread diameter which was measured along two diagonals is found to be 150 mm [15], as demonstrated in Fig. 1.



Fig.1 Measurement of Spread Diameter

2.3 Compressive Strength Test

After mixing the ECC in accordance with [13], place the fresh ECC mix in the cube mould of size 70.6 x 70.6 x 70.6 mm [16]. Then specimens were demolded after 24 hours and wrapped them with wet cloths and stored in moist room until taken out just prior to breaking. After they have been taken out and until they are broken, the cubes shall not be allowed to become dry. Then the cubes were tested for compressive strength in the laboratory using compression testing machine for each curing period [16]. Compressive strength of each specimen is calculated by considering the maximum load applied to the cube. Average of three cube specimens is

considered as ultimate compressive strength for each curing period. The variation in compressive strength with exposure age, as shown in Fig 3, was represented using Matlab.

2.4 Flexural Strength Test

Prism specimens measuring 370 mm × 55 mm × 45 mm were prepared to assess the flexural strength of ECC. After demolding at 24 hours, the specimens were wrapped in wet cloth and stored at room temperature for curing at designated intervals. Flexural tests were conducted [17], and the failure load for each specimen was recorded. The ultimate flexural strength of each trial mix was determined by averaging the results from three identical specimens. The strength was calculated using the formula:

$$\text{Flexural Strength} = \frac{PL}{BD^2}$$

where P represents the applied load in Newtons, L is the span length of the beam in millimeters, B and D are specimen width and depth in millimeters respectively. Flexural strength vs exposure age in days is plotted as shown in Fig. 4.

2.5 Durability Test

To evaluate the durability of ECC in aggressive environments, two solutions were separately prepared: 5% (by mass) sodium sulphate (Na_2SO_4) and 5% (by mass) sodium chloride (NaCl), simulating sulphate and chloride exposures. After 28 days of water curing, ECC specimens were immersed in the respective solutions for 30, 60, 90, 120, 150, 200, 250, 300, 350, and 400 days [18] under laboratory conditions, as shown in Fig. 2. After exposure, the specimens were conditioned at room temperature for 24 hours before being subjected to compression testing.



Fig. 2. Immersion of ECC specimens in solution

The Rapid Chloride Penetration Test (RCPT) was conducted in the laboratory following the guidelines provided in ASTM C-1202 [19] to determine the electrical conductivity of ECC at different curing ages. The total charge passed, measured in coulombs, was calculated during this period. The test setup consisted of two reservoirs, where the specimen was fixed using a bonding agent to ensure a leak-proof seal. One reservoir was filled with a 3% NaCl solution, while the other contained a 0.3N sodium hydroxide solution. A 60V electric current was applied to the specimen, which measured 100 mm in diameter and 50 mm in thickness. The current was recorded at 30-minute intervals over duration of 6 hours. The experimental results were qualitatively assessed and evaluated in accordance with ASTM C-1202 [19-21].

2.6 Microstructural Analysis

To study the pore structure, fiber-matrix bonding and product of hydration, ECC were analysed using Scanning Electron Microscope (SEM). After exposure of ECC cubes at various ages in Water, Sulphate and Chloride environment, specimens were removed and allowed to air dry before cutting in to sections of 5 mm thick [22]. These sections were further trimmed into cuboidal specimens approximately $5 \times 20 \times 20$ mm using a high-speed marble cutter. The specimen surfaces were ground using silicon carbide abrasive papers with grits of 400, 600, and 1,000, followed by polishing with diamond paste at progressively finer grades (6 μm , 3 μm , 1 μm , and 0.25 μm) to achieve the required smoothness for analysis [22]. Tescan Vega 3 SEM, capable of generating Back Scattered Electrons (BSE) and secondary electron (SE) signals, was employed for the examination. Secondary electrons, which result from low-energy inelastic interactions with the specimen, provide detailed information on surface topography due to their emission from the surface or nearby regions. Conversely, backscattered electrons are high-energy electrons that interact elastically with the sample, with their scattering intensity is influenced by the atomic number of the elements present—higher atomic number elements scatter more and appear brighter in the imaging [22, 23].

3. Results and Discussion

3.1 Compressive Strength

Fig. 3 shows the compressive strength development of ECC specimens subjected to water, Na_2SO_4 , and NaCl exposures at different ages. This plot, illustrates the variations in compressive strength from the initial value of 51.88 MPa.

Under water exposure, the compressive strength of ECC specimens exhibits a steady increase. After 30 days, the strength shows a 1.06% gain, reaching 52.43 MPa. By 60 days, the strength rises to 56.92 MPa, reflecting a 9.71% increase. This upward trend continues, with a 12.71% gain at 90 days (58.47 MPa), and a 14.33% increase at 120 days, reaching 59.31 MPa. At 150 days, the compressive strength reaches 59.85 MPa, marking a 15.37% gain. The strength stabilizes after 200 days at 60.12 MPa, showing a 15.90% improvement. After 250 days, the strength slightly increases to 60.58 MPa (16.79% gain), and by 300 days, it reaches 60.81 MPa (17.24% gain). At 350 days, the strength is 61.08 MPa, indicating a 17.75% increase, and after 400 days, it reaches 61.17 MPa, representing an overall gain of 17.90%. This demonstrates a consistent, moderate, improvement in compressive strength under water exposure, with more significant gains observed after approximately 200 days.

Under Na_2SO_4 exposure, ECC exhibits a more pronounced increase in compressive strength over time. After 30 days, the strength rises by 3.53% to 53.71 MPa. At 60 days, the strength increases by 5.68%, reaching 54.83 MPa. The trend continues with a 7.54% gain at 90 days, resulting in a compressive strength of 55.79 MPa. By 120 days, the strength reaches 57.51 MPa, marking a 10.85% improvement. At 150 days, the strength rises to 59.44 MPa, a 14.58% increase, and by 200 days, it reaches 62.76 MPa, showing a significant 20.96% gain. The compressive strength climbs to 63.81 MPa at 250 days, reflecting a 23.02% increase, and further rises to 64.73 MPa (24.76% gain) after 300 days. At 350 days, the strength is 65.61 MPa, indicating a 26.47% improvement. Finally, after 400 days, the ECC exposed to Na_2SO_4 reaches 66.89 MPa, resulting in an overall gain of 28.95%. This indicates that sulphate exposure has a markedly positive effect on ECC strength, continuously enhancing it throughout the exposure period.

In contrast, NaCl exposure leads to a gradual decline in compressive strength over time. After 30 days, the strength drops by 0.81% to 51.46 MPa. At 60 days, the strength decreases further to 50.96 MPa, a 1.77% loss. This downward trend continues at 90 days, where the strength reduces by 2.48% to 50.59 MPa, and by 120 days, it falls to 50.07 MPa, showing a 3.49% loss. At 150 days, the compressive strength decreases by 4.14% to 49.73 MPa, and by 200 days, it reaches 49.35 MPa, marking a 4.87% reduction. At 250 days, the strength drops further to 48.98 MPa, a 5.58% loss, and at 300 days, it decreases to 48.72 MPa, marking a 6.09% reduction. The decline continues at 350 days with strength of 48.07 MPa, showing a 7.34% loss. Finally, after 400 days, the compressive strength reaches 47.83 MPa, representing an overall 7.81% loss. This steady

decline indicates the detrimental impact of chloride exposure on ECC, likely due to chloride ion impetration can cause material degradation over time [24]

Overall, the trends indicate that water exposure leads to moderate and steady gains in compressive strength, stabilizing after 200 days. Whereas in Na_2SO_4 exposure the compressive strength continuously increases showing the enhancement of durability of ECC in sulphate environment [25]. In contrast, NaCl exposure leads to inappreciable reduction in compressive strength of about 7.81 % over 400 days, emphasizing the detrimental effect on ECC in chloride prone environment. The findings of compressive strength test reveal the importance of different environmental conditions while evaluating the long-term durability of ECC.

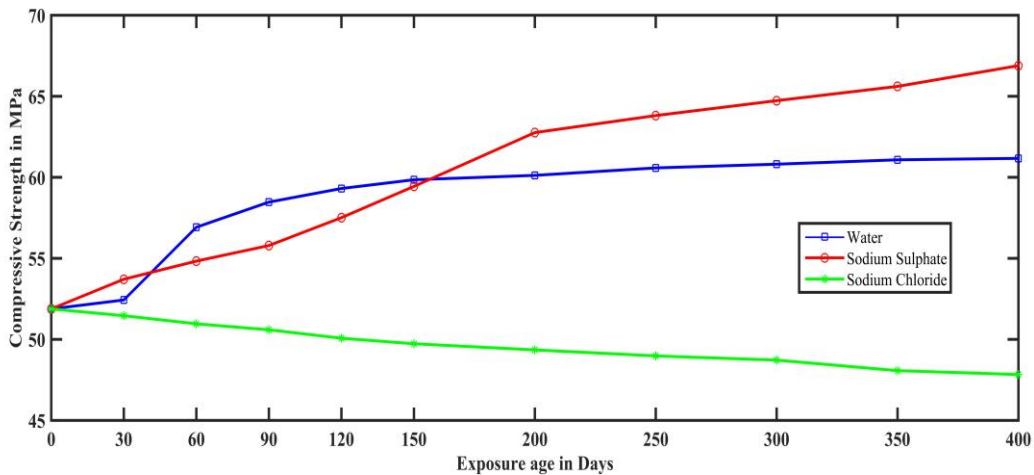


Fig. 3. Compressive Strength of ECC under different exposure

3.2 Flexural Strength

Fig.4 represents the variation of Flexural strength of ECC in Water, Sodium sulphate (Na_2SO_4), and Sodium chloride (NaCl) environment for a period up to 400 days, demonstrating the noticeable trend under chemical environment.

Under water exposure, the flexural strength exhibits a steady and gradual increase over the 400-day period. From an initial strength of 13.76 MPa, the strength rises to 17.39 MPa by the 400th day, marking an overall increase of 26.40%. Incremental growth is particularly evident during the early exposure period (0 to 150 days), during which the strength increases by approximately 17.90% (from 13.76 MPa to 16.22 MPa). Beyond 150 days, the strength continues to improve, albeit at a slower rate, reaching 17.39 MPa at 400 days. This consistent improvement reflects

ongoing hydration and densification of the cementitious matrix in the absence of aggressive chemical interactions [26]

Under Na_2SO_4 exposure, the ECC specimens demonstrate superior flexural strength performance compared to those exposed to water. The strength increases from 13.76 MPa initially to 18.73 MPa at 400 days, representing a remarkable total increase of 36.1%. Early-stage increments (0 to 150 days) are significant, with strength rising by 22.5% (from 13.76 MPa to 16.86 MPa). Beyond 150 days, the increase becomes even more pronounced, with a growth of 11.1% between 200 and 400 days (from 17.17 MPa to 18.73 MPa). This enhanced performance is likely due to secondary ettringite formation induced by sulphate ions, which contributes to matrix densification and improved load-bearing capacity [25].

Conversely, NaCl exposure results in a continuous decline in flexural strength, highlighting the detrimental impact of chloride ions. The strength decreases from 13.76 MPa initially to 10.82 MPa at 400 days, representing a total reduction of 21.4%. The decline is most pronounced during the early exposure period (0 to 150 days), with strength decreasing by 8.5% (from 13.76 MPa to 12.59 MPa). Beyond 150 days, the reduction persists, with a further decrease of 14% (from 12.17 MPa at 200 days to 10.82 MPa at 400 days). This degradation can be attributed to chloride-induced disruption of hydration products and the potential initiation of micro-cracking, which weakens the composite structure [24]

Compared to the water-exposed specimens, the Na_2SO_4 exposed specimens exhibit consistently higher flexural strength throughout the exposure period, with an average increase of 8% to 10% across all intervals. In contrast, NaCl -exposed specimens show significantly lower strength than the control specimens, with reductions ranging from 6% at 30 days to nearly 37.7% by the end of the 400-day period. These findings highlight the beneficial effects of sulphate exposure in enhancing ECC durability, whereas chloride exposure poses a substantial risk to its long-term mechanical performance [25, 26]. The results provide critical insights into the suitability of ECC for environments with varying chemical exposures.

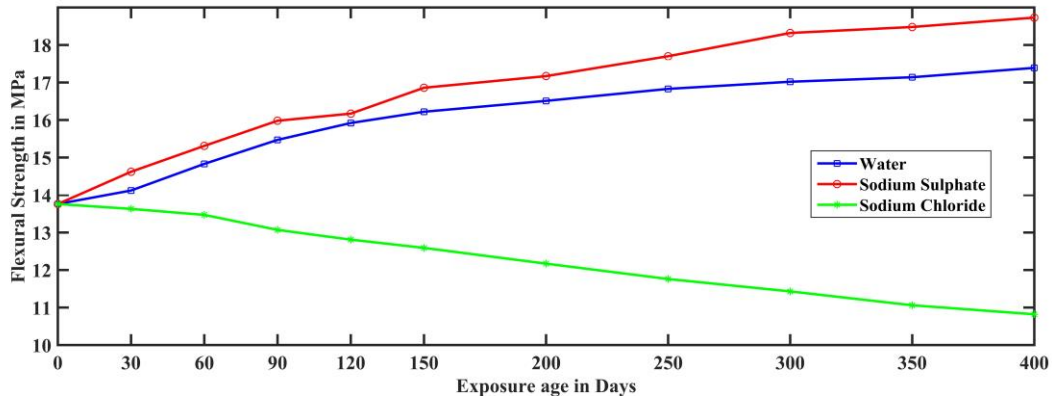


Fig. 4. Flexural Strength of ECC under different exposure

3.3 RCPT Results



Fig. 5 RCPT Test Setup

RCPT measures the resistivity of ECC to chloride ion penetration. A specimen with a diameter of 100 mm was tested for electrical conductance using the RCPT setup, as shown in Fig. 5.

The experimental results for the charge passed through ECC specimens, measured in coulombs over a 400-day period, are presented in Table 2. These results reveal a distinct trend of decreasing charge passed as the material ages, indicative of improved resistance to chloride ion penetration over time. Initially, at 30 days, the charge passed is 1350 coulombs. This value progressively decreases to 1322 coulombs at 60 days and 1256 coulombs at 90 days. The decline continues more substantially, with the charge passed reducing to 1118 coulombs by 120 days and further to 1090 coulombs at 150 days. The rate of decrease slows as the specimens age, with values of 1066 coulombs at 200 days, 1063 coulombs at 250 days, and 1057 coulombs at 300 days. By the end of the 400-day exposure period, the charge passed reaches its lowest value of 1020 coulombs. This consistent reduction in charge passed highlights the progressive densification of the ECC matrix over time. This improvement is attributed to ongoing hydration and pozzolanic reactions, particularly involving the supplementary cementitious materials like

fly ash and GGBS. The observed decrease in porosity and the increased formation of calcium-silicate-hydrate (C-S-H) phases are likely contributing to the enhanced durability and reduced permeability of the composite [27]. This results in a more effective barrier against the ingress of aggressive ions such as chlorides. The overall trend demonstrates that ECC exhibits excellent long-term performance with respect to chloride ion penetration resistance, as evidenced by the decreasing coulomb values, which remain well below the 2000 coulombs threshold specified by ASTM C1202 [19] for low permeability. The results of RCPT indicates the suitability of ECC in marine, coastal infrastructure or areas where structure exposed to chloride environment, thereby enhancing its potential use in chloride rich environment.

Table 2 RCPT values of ECC at different age

Age of ECC in Days	Charge Passed in Coulombs
30	1350
60	1322
90	1256
120	1118
150	1090
200	1066
250	1063
300	1057
350	1048
400	1020

3.4 Curve Fitting

Curve fitting tool of MATLAB is a flexible and robust software, which designed to fit mathematical models to the experimental data. Software has got extensive curve fitting techniques which enables the researcher to analyse the relationships between variables and to predict the trend beyond the collected data sets. The tool also gives the real time monitoring and analysis thereby increasing the accuracy of curve fitting and generates the statistical outputs like root mean square error (RMSE), coefficient of determination (R^2) and residual plots which gives the ideal fit of the curve. Based on the experimental data sets, researchers can select the various polynomial, logarithmic, exponential and custom-made models in order to represent the datasets more suitably [28].

In the present study, the variation of compressive and flexural strength of ECC were analysed, and cubic polynomial fitting was found to be the best method to represent the output trends. A cubic polynomial is expressed as:

$$y = a_3x^3 + a_2x^2 + a_1x + a_0$$

where, y represents the Compression/flexural strength in MPa, x is the exposure age in days, and a_3 , a_2 , a_1 , and a_0 are the coefficients determined during curve fitting. The cubic polynomial method effectively represents gradual variation of strength and any inflection points within the experimental data sets and this method particularly suitable for nonlinear data sets with complex trends.

The compressive strength of ECC under water, sodium sulphate (Na_2SO_4), and sodium chloride (NaCl) exposure conditions are presented in Table.4 shows a noticeable trend up to 400 days for different exposure conditions. The values within the parentheses represents the predicted values generated using MATLAB curve fitting equations. For water exposure, Compressive strength gradually increases from 51.88 MPa to 61.17 MPa and predicted values shows a maximum deviation of only 1.3 %, relative to the experimental values.

For Na_2SO_4 exposure, compressive strength increases significantly from 51.88 MPa to 66.89 MPa, demonstrating the beneficial effects of sulphate ions [18]. The curve-fitting model accurately captures this growth, with deviations generally remaining below 1%. Conversely, there is a gradual decrease of compressive strength from 51.88 MPa to 47.83 MPa was observed that may be due to chloride induced deterioration [30]. In this also curve fitting equation align well with the experimental values, with deviations not exceeding 0.2 MPa.

Table 5 shows a comparison of Flexural Strength experimental results with predicted values. For water exposure, the predicted values show same trend as experimental values, gradually increasing from 13.6 MPa to 17.4 MPa, with a maximum difference observed as 0.3 MPa. The strong correlation between experimental and predicted values justifies the accuracy of curve fitting method. Under Na_2SO_4 exposure, the predicted values increase from 13.8 MPa to 18.8 MPa up to 400 days. The deviation of predicted values from experimental values was below 0.2 MPa, showing the indicating the method's ability which reflects the strength gain of ECC in sulphate exposure. The orientation of predicted values with experimental results further validates the correctness of curve fitting method for composites.

On the other hand, NaCl exposure exhibited decrease in predicted strength values from 13.8 MPa to 10.8 MPa which resembling the experimental results. Here also the maximum variation

of predicted values was limited to 0.3 MPa, reaffirming the accuracy of cubic fit even in scenarios involving material degradation.

For all the exposure conditions the deviation of predicted values is minimum, this emphasizes the robustness of curve fitting method. Also curve fitting method yields high R^2 value of all the environment shows an excellent relationship between both the values [29].

Table 4 Experimental and predicted values Compressive strength of ECC under different exposure condition

Exposure age in Days	Compressive Strength in MPa`		
	Water Exposure	Na ₂ SO ₄ Exposure	NaCl Exposure
0	51.88 (51.2)	51.88 (51.8)	51.88 (51.9)
30	52.43 (54.1)	53.71 (53.3)	51.46 (51.4)
60	56.92 (56.3)	54.83 (54.9)	50.96 (50.9)
90	58.47 (57.9)	55.79 (56.4)	50.59 (50.5)
120	59.31 (59.1)	57.51 (57.9)	50.07 (50.1)
150	59.85 (59.9)	59.44 (59.4)	49.73 (49.8)
200	60.12 (60.6)	62.76 (61.7)	49.35 (49.4)
250	60.58 (60.7)	63.81 (63.6)	48.98 (49.00)
300	60.81 (60.7)	64.73 (65.2)	48.72 (48.6)
350	61.08 (60.8)	65.61 (66.2)	48.07 (48.2)
400	61.17 (61.3)	66.89 (66.5)	47.83 (47.8)

Table 5 Experimental and predicted values Flexural strength of ECC under different exposure condition

Exposure age in Days	Flexural Strength in MPa		
	Water Exposure	Na ₂ SO ₄ Exposure	NaCl Exposure
0	13.76 (13.6)	13.76 (13.8)	13.76 (13.8)
30	14.12 (14.3)	14.62 (14.6)	13.63 (13.6)
60	14.83 (14.9)	15.31 (15.3)	13.47 (13.4)
90	15.47 (15.4)	15.98 (15.8)	13.07 (13.1)
120	15.92 (15.8)	16.17 (16.3)	12.81 (12.9)
150	16.22 (16.2)	16.86 (16.8)	12.59 (12.6)
200	16.51 (16.6)	17.17 (17.3)	12.17 (12.2)
250	16.83 (16.8)	17.7 (17.8)	11.76 (11.8)
300	17.02 (17)	18.32 (18.1)	11.43 (11.4)
350	17.14 (17.2)	18.48 (18.5)	11.06 (11.1)
400	17.39 (17.4)	18.73 (18.8)	10.82 (10.8)

*The values with in the parenthesis represent the value of strength predicted by curve fitting equation.

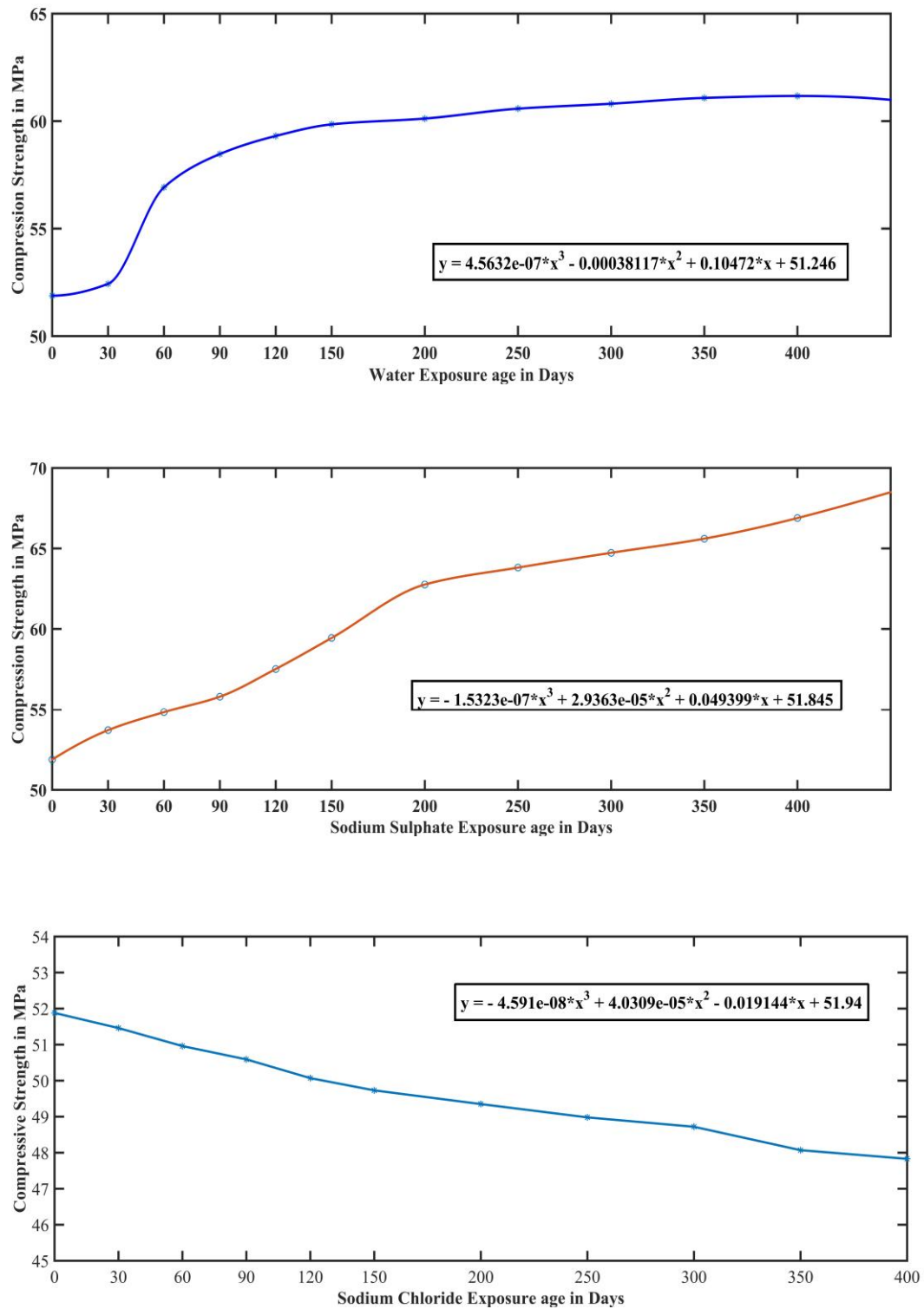


Fig. 6 Compressive strength Curve Fitting under water, Sodium Sulphate and Sodium Chloride Exposure along with the Cubic polynomial equation for prediction of Compressive strength

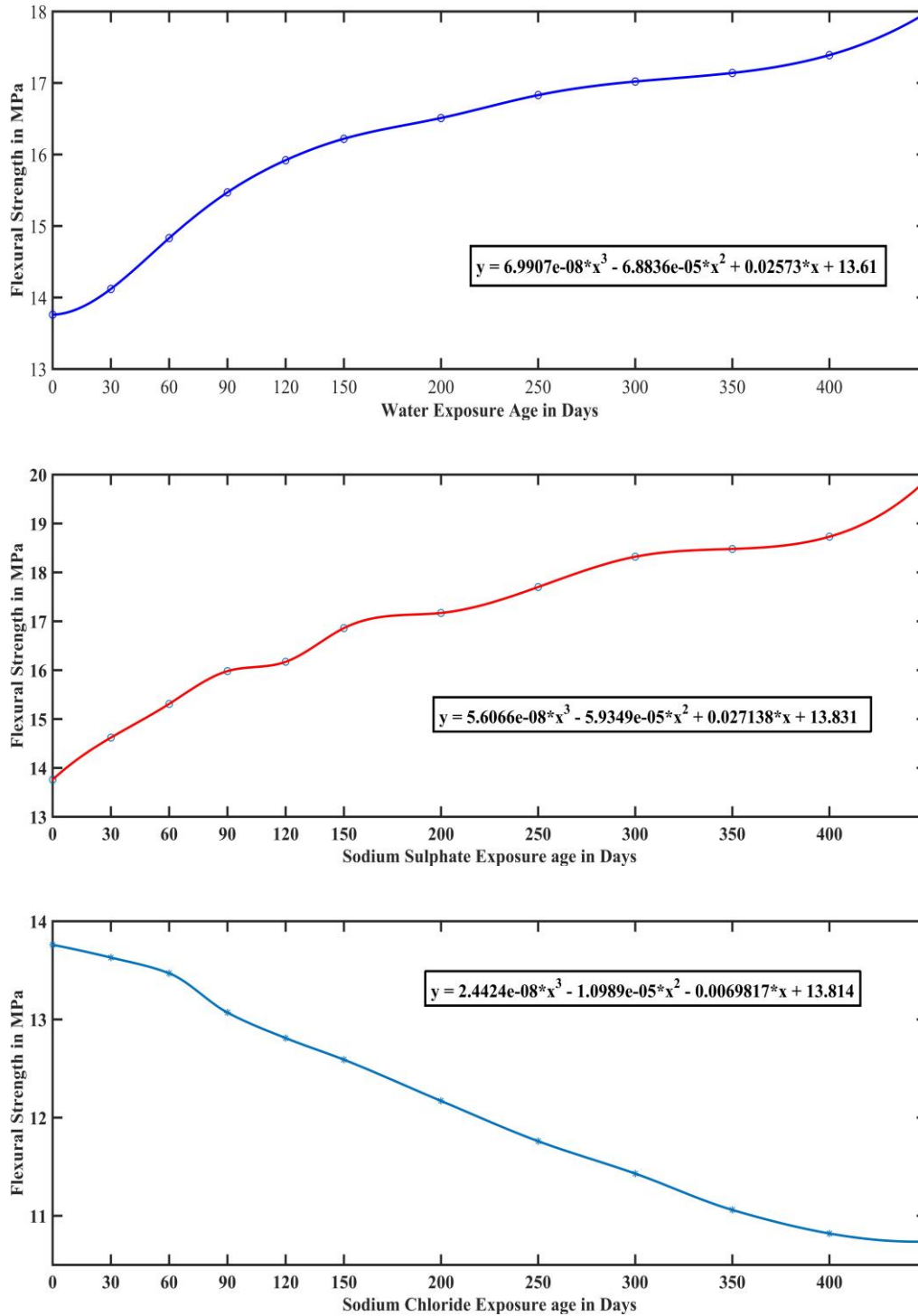


Fig. 7 Flexural strength Curve Fitting under water, Sodium Sulphate and Sodium Chloride Exposure along with the Cubic polynomial equation for prediction of Flexural strength

The compressive strength trends of ECC under water, Na_2SO_4 , and NaCl exposure conditions, as depicted in Fig. 6, were effectively modeled using MATLAB's curve-fitting tool, yielding three cubic polynomial equations with R^2 values ranging from 0.9935 to 0.9979. Similarly, the flexural strength of ECC under water, Na_2SO_4 , and NaCl exposure conditions, shown in Fig. 7, was also

successfully modeled using cubic polynomial equations, with R^2 values ranging from 0.9908 to 0.9967. The predicted values demonstrated exceptional alignment with experimental results, with deviations consistently below 0.3 MPa, reinforcing the model's capability to capture significant variations in flexural strength.

These high R^2 values indicating the absolute agreement between the predicted and experimental values [28, 29], justifying the reliability of the fitted methods. Overall, the MATLAB curve-fitting tool demonstrates exceptional accuracy in modeling both strength gains and losses, providing a appropriate solution for understanding ECC behavior under different environmental conditions. The minimal deviations between predicted and experimental values authenticates the applicability of the cubic polynomial expression for accurately predicting long-term strength trends.

3.5 Microstructural Studies

To investigate the microstructure of ECC subjected to water, sulphate, and chloride exposure, a SEM study was conducted. The analysis of ECC microstructures exposed to water at various intervals is illustrated in Fig. 8. Initially, ECC cured for 28 days in water exhibits a dense and compact microstructure, characterized by a well-formed calcium silicate hydrate (C-S-H) matrix and minimal porosity, indicating effective hydration promoted by water curing. Fig 8.a reveals the presence of unreacted fly ash and supplementary cementitious materials such as GGBS, which contribute to further matrix densification. This observation suggests enhanced long-term durability and improved resistance to environmental degradation, including chloride and sulphate attacks. Additionally, Fig. 8(a) shows the bond between the fibers and the matrix, with the structural integrity appearing robust, minimal microcracking, and effective early-age crack control, due to the reinforcement provided by the fibers. In contrast, Fig. 8(b) depicts the SEM image of ECC after 90 days in water, where some residual fly ash and GGBS particles are observed, indicating the continuous hydration of pozzolanic materials. [31]. The presence of micro voids, surface pitting, and cracks suggests that prolonged exposure to water introduces internal stresses that weakens the composite structure over time [32].

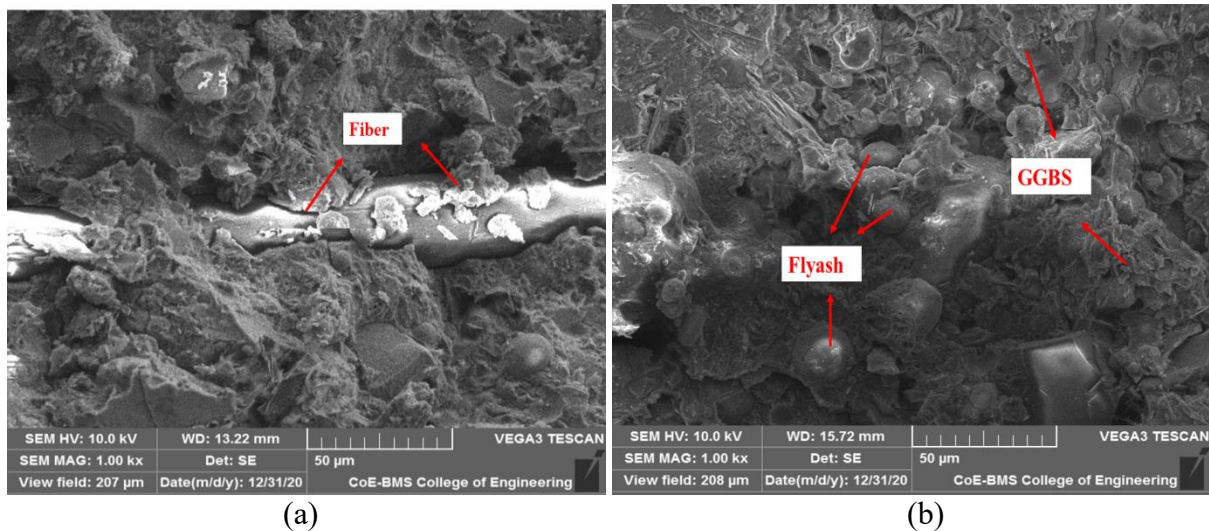


Fig. 8 SEM images of ECC (a) after 28 days in water, (b) after 90 days in water

Fig. 9 provides a comprehensive analysis of the microstructural behaviour of ECC exposed to a Na_2SO_4 environment over prolonged durations, offering critical insights into the effect of sulphate ions. The SEM image of ECC after 120 days of immersion in Na_2SO_4 , shown in Fig 9.a, reveals significant microstructural alterations associated with extended sulphate exposure. The magnified surface morphology illustrates a densely compacted matrix with visible formations of fibrous and interconnected micro-pores. This compactness suggests enhanced hydration products, likely calcium silicate hydrates (C-S-H), contributing to the material's densification over time. Further analysis of the ECC sample cured for 200 days in Na_2SO_4 , as depicted in Fig 9.b, demonstrates the absence of expansive compounds such as ettringite, which are typically associated with expansive behaviour within the matrix. This absence attributes the strength gain even after 200 days of exposure. Presence of densified regions with compact micro cracking indicates that, ECC shows good bonding and durability under long term exposure to sulphate with some internal effect of micro cracking. But this micro cracking does not affect the overall performance of ECC under such exposure condition. The findings emphasize the material durability and good bonding between the matrix and fiber even after prolonged exposure to sulphate environment.

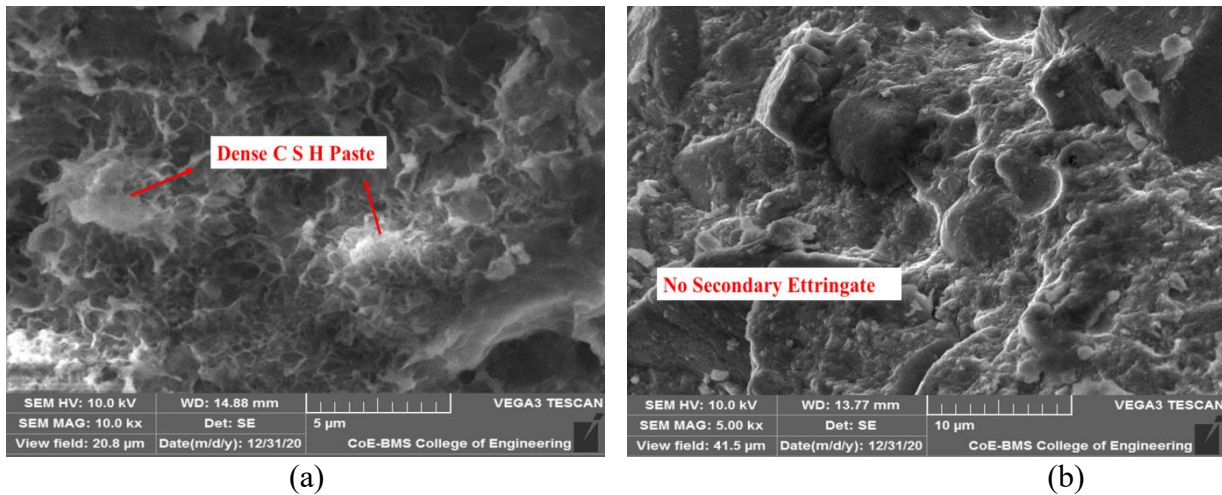


Fig. 9 SEM images of ECC (a) after 120 days in Na_2SO_4 (b) after 200 days in Na_2SO_4

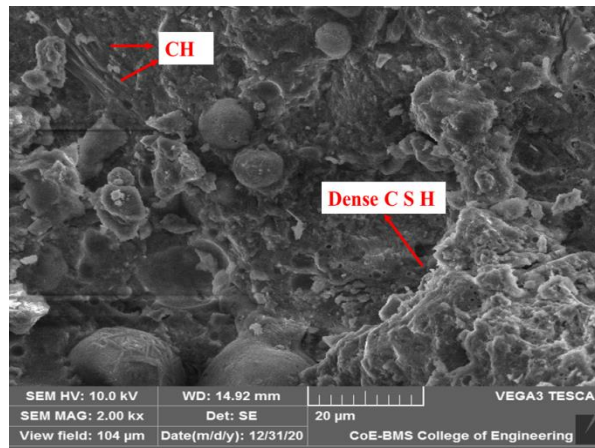


Fig. 10 SEM images of ECC after 120 days in NaCl

The SEM image of ECC exposed to a NaCl solution for 120 days, as shown in Fig. 10, reveals significant microstructural deterioration. Even there is presence of dense CSH gel, the strength gradually decreases because of micro pores and cracks. This cracks and pores might affect the material integrity and strength and it is observed that maximum decrease in compressive strength and flexural strength is 7 % and 21.36 % respectively. Also, SEM image reveals the presence of degraded region, which ensures the penetration of chloride ions through pores and cracks making ECC susceptible to the chloride induced deterioration [26]. Hence, there is a necessity to improve the resistance of materials in composites, when ECC is subjected to long term exposure to chloride environment.

4. Conclusions

The study of ECC mechanical behaviour under water, sodium sulphate (Na_2SO_4), and sodium chloride (NaCl) exposure over 400 days provides critical insights into its long-term durability,

with both compressive and flexural strength trends revealing distinct effects of environmental conditions. Under water exposure, ECC demonstrated a steady increase in compressive strength, stabilizing around 60 MPa after 200 days, indicating enhanced long-term hydration and matrix densification, despite the presence of micro voids and cracks at later ages, which got minimal impact on strength due to controlled crack widths [33]. Flexural strength also followed similar trend, increasing from 13.76 MPa to 17.39 MPa up to 400 days, with gradual increase in strength was observed up to 150 days thereafter the strength gain slowed down indicating the role of continued hydration in strength enhancement. The most significant strength increase was observed under Na_2SO_4 exposure, where maximum compressive and flexural strength are 66.89 MPa and 18.73 MPa, respectively. This strength increase is attributed by secondary reaction, which contributed to matrix densification and increased load-bearing capacity without inducing surface cracking [18]. This improvement, further supported by ECC's superior tensile properties, delayed cracking and restricted the ingress of aggressive ions. Additionally, no surface cracks or delayed ettringite formation were observed in sulphate-exposed specimens, further stressing the importance of ECC resistance to sulphate attack. Conversely, NaCl exposure led to an initial increase in compressive and flexural strength, followed by a gradual decline to 50 MPa and 10.82 MPa, respectively, by day 400, likely due to chloride-induced degradation and potential fiber corrosion [30]. SEM images confirmed the presence of fly ash and GGBS in the ECC mix, which reduced heat of hydration, improved workability, and facilitated prolonged hydration, enhancing resistance to sulphate and chloride penetration. The Rapid Chloride Permeability Test (RCPT) indicated low chloride ion penetrability, with charges passed consistently below 2000 coulombs, meeting ASTM-C-1202 standards for low permeability [30]. The cubic polynomial equations generated through MATLAB curve fitting method, with R^2 values ranging from 0.9935 to 0.9979 for compressive strength and 0.9908 to 0.9967 for flexural strength. Also, MATLAB predicted values aligns closely with experimental results and deviations consistently below 0.3 MPa, confirming the robustness of the cubic polynomial method for prediction of strength. The high R^2 values of curve fitting equation for all exposure condition revealed the applicability of curve fitting method for predicting ECC long-term performance and also provides the valuable insights into its durability under aggressive environmental conditions. While sulphate exposure enhances ECC mechanical performance, chloride exposure demands careful attention while studying long-term durability due to potential microstructural degradation [24, 25]. These findings emphasize the ECC suitability for infrastructure applications in sulphate-rich

environments, while highlighting the need for protective measures against chloride-induced deterioration.

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Statements & Declarations

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Competing Interests

The authors declare no conflict of interest.